1. To write a Python program to add two numbers entered by the user.

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

sum_numbers = num1 + num2

print(f"The sum of {num1} and {num2} is: {sum_numbers}")
```

```
Enter the first number: 5
Enter the second number: 7
The sum of 5.0 and 7.0 is: 12.0
```

2. To write a Python program to interchange the values of two variables.

```
var1 = input("Enter the value for variable 1: ")
var2 = input("Enter the value for variable 2: ")

print(f"\nBefore swapping:")
print(f"Variable 1: {var1}")
print(f"Variable 2: {var2}")

var1, var2 = var2, var1

print(f"\nAfter swapping:")
print(f"Variable 1: {var1}")
print(f"Variable 2: {var2}")
```

```
Enter the value for variable 1: 2
Enter the value for variable 2: 2

Before swapping:
Variable 1: 2
Variable 2: 2

After swapping:
Variable 1: 2
Variable 2: 2
```

3. To write a Python program to find the largest among three numbers using conditional statements

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))
```

```
if (num1 >= num2) and (num1 >= num3):
   largest = num1
elif (num2 >= num1) and (num2 >= num3):
   largest = num2
else:
   largest = num3

print(f"The largest number among {num1}, {num2}, and {num3} is: {largest}")
```

```
Enter the first number: 2
Enter the second number: 2
Enter the third number: 2
The largest number among 2.0, 2.0, and 2.0 is: 2.0
```

4. To write a Python program to check whether a given number is even or odd.

```
num = int(input("Enter an integer: "))

if (num % 2) == 0:
   print(f"{num} is an even number.")
else:
   print(f"{num} is an odd number.")
```

```
Enter an integer: 2
2 is an even number.
```

5. To write a Python program to determine whether a number is positive, negative, or zero.

```
num = float(input("Enter a number: "))

if num > 0:
   print(f"{num} is a positive number.")
elif num < 0:
   print(f"{num} is a negative number.")
else:
   print(f"{num} is zero.")
```

```
Enter a number: 3
3.0 is a positive number.
```

6. To write a Python program to check whether a given year is a leap year.

```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
```

```
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

```
Enter a year: 2006
2006 is not a leap year.
```

7. To write a Python program to perform basic arithmetic operations using operators.

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
print(f"{num1} + {num2} = {num1 + num2}")


print(f"{num1} - {num2} = {num1 - num2}")


print(f"{num1} * {num2} = {num1 * num2}")

if num2 != 0:
    print(f"{num1} / {num2} = {num1 / num2}")
else:
    print("Cannot divide by zero!")
```

```
Enter the first number: 2
Enter the second number: 2
2.0 + 2.0 = 4.0
2.0 - 2.0 = 0.0
2.0 * 2.0 = 4.0
2.0 / 2.0 = 1.0
```

8. To write a Python program to display the multiplication table of a given number.

```
num = int(input("Enter a number to see its multiplication table: "))

print(f"Multiplication Table for {num}:")
for i in range(1, 11): # Loop from 1 to 10
    print(f"{num} x {i} = {num * i}")
```

```
Enter a number to see its multiplication table: 12
Multiplication Table for 12:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
```

```
12 x 9 = 108
12 x 10 = 120
```

9. To write a Python program to calculate the factorial of a number using a loop.

```
num = int(input("Enter a non-negative integer: "))

factorial = 1

# Check if the number is negative, zero, or positive
if num < 0:
   print("Factorial does not exist for negative numbers")
elif num == 0:
   print("The factorial of 0 is 1")
else:
   for i in range(1, num + 1):
       factorial = factorial * i
   print(f"The factorial of {num} is {factorial}")
```

```
Enter a non-negative integer: -6
Factorial does not exist for negative numbers
```

10.To write a Python program to find the sum of the first N natural numbers.

```
n = int(input("Enter a positive integer N: "))

if n < 0:
    print("Please enter a positive integer.")
else:
    sum_n = 0
    for i in range(1, n + 1):
        sum_n += i
    print(f"The sum of the first {n} natural numbers is: {sum_n}")
```

```
Enter a positive integer N: 4
The sum of the first 4 natural numbers is: 10
```

11.To write a Python program to print numbers from 1 to N using a for loop.

```
n = int(input("Enter a positive integer N: "))

if n <= 0:
    print("Please enter a positive integer.")
else:
    print(f"Numbers from 1 to {n}:")
    for i in range(1, n + 1):
        print(i)
```

```
Enter a positive integer N: 3
Numbers from 1 to 3:
1
2
3
```

12.To write a Python program to print numbers in reverse order from N to 1.

```
n = int(input("Enter a positive integer N: "))

if n <= 0:
    print("Please enter a positive integer.")
else:
    print(f"Numbers from {n} down to 1:")
    for i in range(n, 0, -1):
        print(i)
```

```
Enter a positive integer N: 3
Numbers from 3 down to 1:
3
2
1
```

13. To write a Python program to print all even numbers up to a given limit.

```
limit = int(input("Enter the limit: "))

if limit < 0:
    print("Please enter a non-negative limit.")
else:
    print(f"Even numbers up to {limit}:")
    for i in range(1, limit + 1):
        if i % 2 == 0:
            print(i)
```

```
Enter the limit: 2
Even numbers up to 2:
2
```

14. To write a Python program to reverse the digits of a given number.

```
num = int(input("Enter an integer: "))

reversed_num = int(str(abs(num))[::-1])

if num < 0:
    reversed_num = -reversed_num
```

```
print(f"The reversed number is: {reversed_num}")
```

```
Enter an integer: 5
The reversed number is: 5
```

15. To write a Python program to count the number of digits in a given integer.

```
num = int(input("Enter an integer: "))

digit_count = len(str(abs(num)))

print(f"The number of digits in {num} is: {digit_count}")
```

```
Enter an integer: 4
The number of digits in 4 is: 1
```

16.To write a Python program to calculate the sum of digits of a given number.

```
num = int(input("Enter an integer: "))

temp_num = abs(num)
sum_of_digits = 0

while temp_num > 0:
    digit = temp_num % 10
    sum_of_digits += digit
    temp_num //= 10

print(f"The sum of the digits of {num} is: {sum_of_digits}")
```

```
Enter an integer: 34
The sum of the digits of 34 is: 7
```

17.To write a Python program to check whether a given number is prime.

```
num = int(input("Enter an integer: "))

is_prime = True

if num <= 1:
    is_prime = False
elif num == 2:
    is_prime = True
else:
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
```

```
            break

if is_prime:
    print(f"{num} is a prime number.")
else:
    print(f"{num} is not a prime number.")
```

```
Enter an integer: 67
67 is a prime number.
```

18.To write a Python program to generate the Fibonacci series up to N terms.

```
n_terms = int(input("Enter the number of terms (N) for the Fibonacci series: ")

a, b = 0, 1
count = 0

if n_terms <= 0:
    print("Please enter a positive integer.")
elif n_terms == 1:
    print("Fibonacci series:")
    print(a)
else:
    print("Fibonacci series:")
    while count < n_terms:
        print(a, end=" ")
        nth = a + b
        a = b
        b = nth
        count += 1
```

```
Enter the number of terms (N) for the Fibonacci series: 98
Fibonacci series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711
```

19.To write a Python program to check whether a string is a palindrome.

```
text = input("Enter a string: ")

# Remove spaces and convert to lowercase for case-insensitive and space-agnosti
cleaned_text = text.replace(" ", "").lower()

# Check if the cleaned string is equal to its reverse
if cleaned_text == cleaned_text[::-1]:
    print(f"'{text}' is a palindrome.")
else:
    print(f"'{text}' is not a palindrome.")
```

```
Enter a string: 9
'9' is a palindrome.
```

20.To write a Python program to check whether a given number is an Armstrong number.

```
num = int(input("Enter an integer: "))

original_num = abs(num)

num_str = str(original_num)
num_digits = len(num_str)

sum_of_powers = 0


for digit_char in num_str:
    digit = int(digit_char)
    sum_of_powers += digit ** num_digits

if sum_of_powers == original_num:
    print(f"{num} is an Armstrong number.")
else:
    print(f"{num} is not an Armstrong number.")
```

```
Enter an integer: 9
9 is an Armstrong number.
```